

GDM Risk Predictor — EDA Report

Dataset: dataset_en.csv | Rows: 10000 | Features: 6

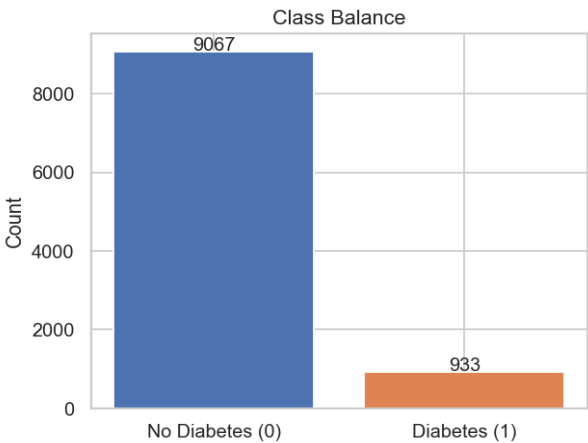
1. Descriptive Statistics

	Age	Previous_Pregnancies	Previous_C_Sections	Diastolic_BP_mmHg	Serum_Insulin_2h	BMI
count	10000.0	10000.0	10000.0	10000.0	10000.0	10000.0
mean	30.54	0.96	0.13	71.97	98.22	26.88
std	5.4	1.04	0.37	8.59	50.64	5.07
min	16.0	0.0	0.0	48.0	12.0	17.0
25%	27.0	0.0	0.0	66.0	62.38	23.3
50%	31.0	1.0	0.0	72.0	87.1	26.8
75%	34.0	2.0	0.0	78.0	121.1	30.3
max	46.0	7.0	3.0	105.0	320.0	44.5

2. Class Balance

No Diabetes (0): 9067 (90.7%) | Diabetes (1): 933 (9.3%)

■ Significant class imbalance (91/9). Model will use class_weight='balanced'.

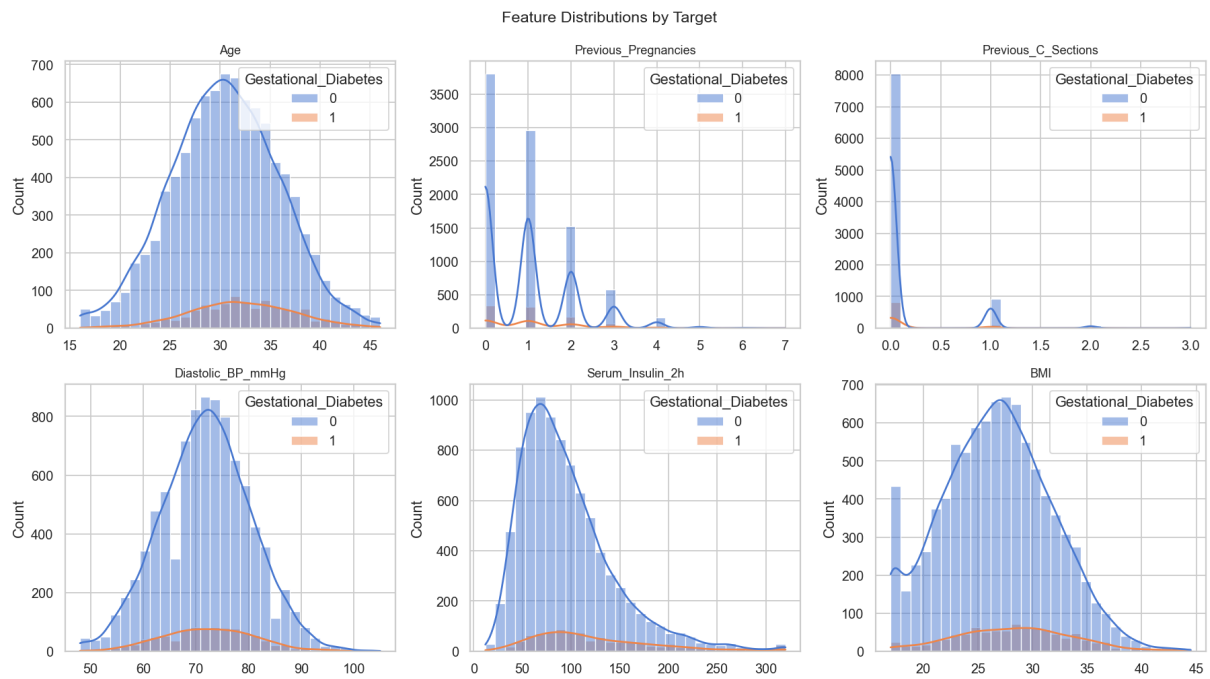


3. Outlier Analysis (IQR Method)

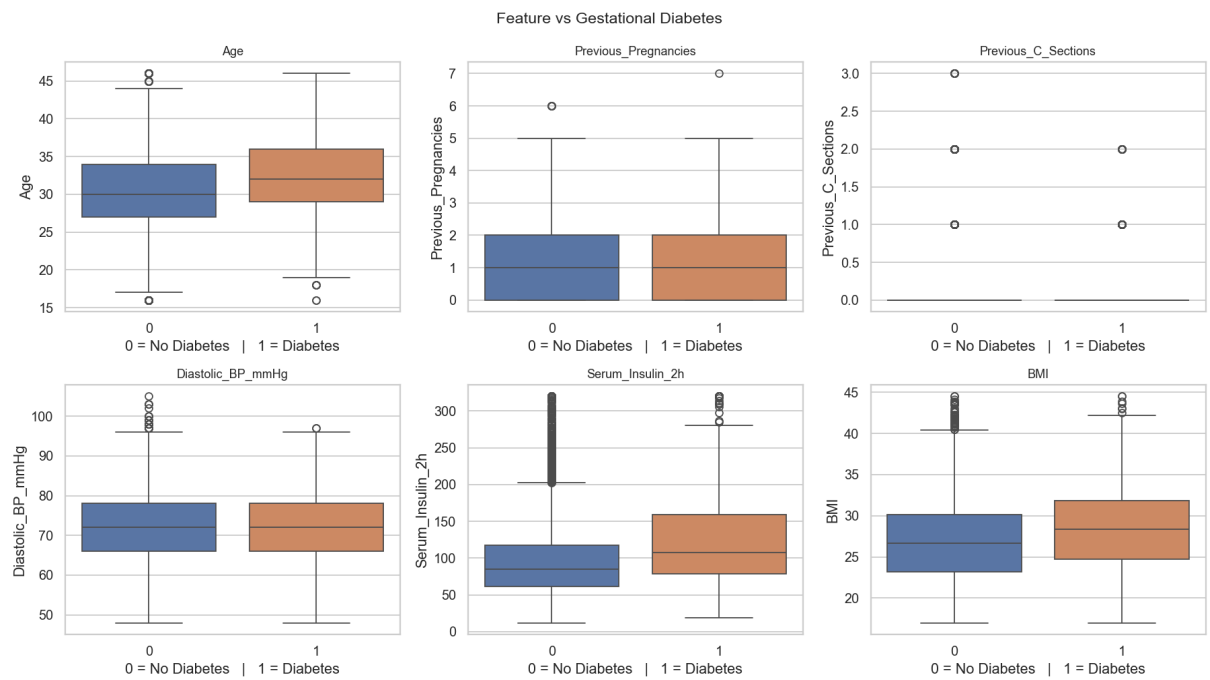
Feature	Q1	Q3	IQR	Fence Low	Fence High	Min	Max	Outliers
Age	27.0	34.0	7.0	16.5	44.5	16.0	46.0	91
Previous_Pregnancies	0.0	2.0	2.0	-3.0	5.0	0.0	7.0	7
Previous_C_Sections	0.0	0.0	0.0	0.0	0.0	0.0	3.0	1137
Diastolic_BP_mmHg	66.0	78.0	12.0	48.0	96.0	48.0	105.0	26
Serum_Insulin_2h	62.4	121.1	58.7	-25.7	209.2	12.0	320.0	414
BMI	23.3	30.3	7.0	12.8	40.8	17.0	44.5	44

Note: Serum_Insulin_2h capped at 320.0 — treated as sensor limit, replaced with median.

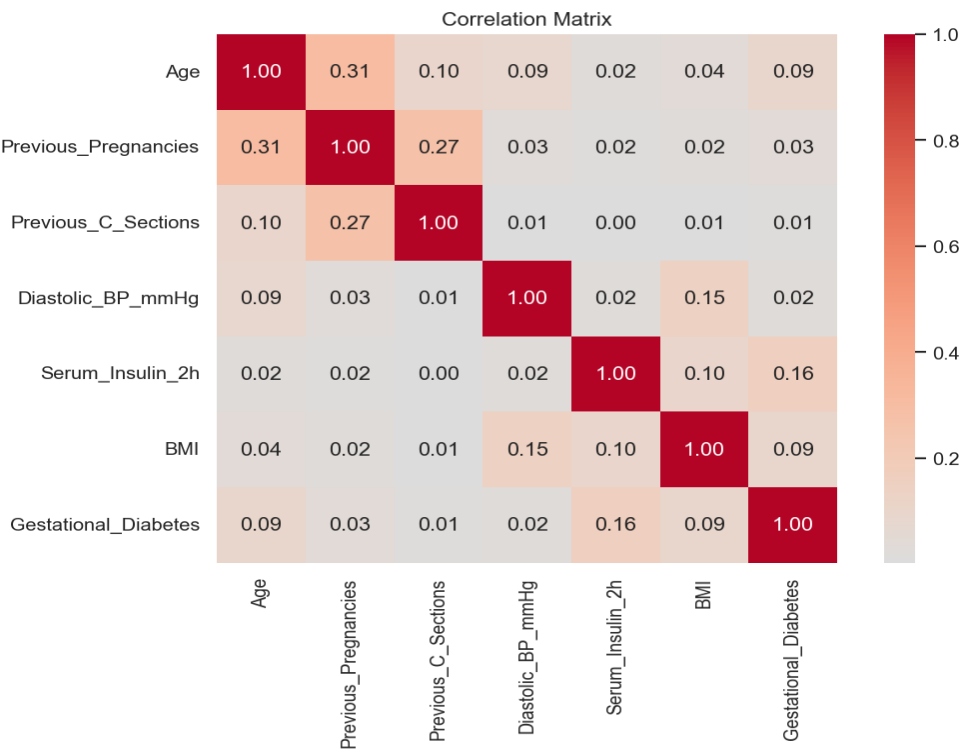
4. Feature Distributions



5. Boxplots by Target



6. Correlation Matrix



Correlations with target (Gestational_Diabetes):

Serum_Insulin_2h: 0.165
Age: 0.092
BMI: 0.090
Previous_Pregnancies: 0.035
Diastolic_BP_mmHg: 0.018
Previous_C_Sections: 0.014

Key insight: Serum_Insulin (0.15) and BMI (0.09) are strongest predictors. Low overall correlations suggest tree-based or regularized models needed.